

Aymen Amimoussa

Data Engineer

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Profile

Data Engineer working across Python, SQL, and PySpark to build ETL/ELT pipelines and Lakehouse architectures (Bronze/Silver/Gold) on Azure and AWS. Comfortable owning incremental load logic, MERGE upserts, and SCD Type 2 modeling, running inside Docker-based environments backed by PostgreSQL. A Machine Learning background adds a useful angle on preprocessing and handling large datasets.

Technical Skills

- **Programming & Querying:** Python, SQL (joins, CTEs, window functions, MERGE), Git
- **Data Engineering:** ETL/ELT pipelines, incremental loads (watermark/timestamp), data modeling, SCD Type 1 & 2, deduplication
- **Big Data & Storage:** Apache Spark (PySpark, Spark SQL), Delta Lake, Parquet
- **Cloud & Platforms:** Azure (Data Factory, ADLS Gen2, Azure SQL Database, Databricks); AWS for storage, processing, and orchestration
- **Infrastructure / DevOps:** Docker, Linux, Azure DevOps (CI/CD basics), GitHub
- **Databases:** PostgreSQL, Azure SQL Database, SQL Server
- **Supporting:** data preprocessing pipelines, automation, dataset handling for analytics and ML

Professional Experience

Data Engineer

Technology Services Company

2026 – Present

Budapest, Hungary

- Build and maintain ETL/ELT pipelines in Python, SQL, and PySpark.
- Use Azure and AWS for data storage, processing, and pipeline orchestration.
- Manage Docker-based infrastructure and the PostgreSQL databases behind it.
- Keep code clean and tested (pylint, pytest) and work through Git-based reviews.

AI Software Developer Intern

mvg Plaut Hungary

Jun 2024 – Sep 2024

Budapest, Hungary

- Built and maintained pipelines for ingesting, preprocessing, and validating industrial data.
- Designed structured datasets and tuned SQL queries for analytics, balancing performance and readability.
- Handled high-volume structured and semi-structured data in workflows close to production.
- Added data quality checks for consistency and completeness, and automated the processing to keep it reproducible.

Database Administrator Intern

Directorate of Ouargla Province

Feb 2023 – Mar 2023

Algeria

- Maintained SQL Server databases (backup, restore, monitoring) and improved performance through indexing.

Database Administrator Intern

Mother & Child Maternity Hospital

Mar 2022 – May 2022

Algeria

- Supported database maintenance and helped streamline data access workflows with the IT team.

Selected Projects

Azure Lakehouse: End-to-End Data Engineering Project

- Designed a Bronze/Silver/Gold Lakehouse architecture on Azure.

- Built ADF pipelines for ingestion and staging, then transformed the data with Databricks and PySpark.
- Implemented incremental loads with watermark/timestamp logic and applied MERGE upserts plus SCD Type 2.
- Delivered a pipeline solid enough for real analytics consumption, not just a proof of concept.

SQL Analytics & Data Modeling

- Worked through joins, CTEs, window functions, top-N per group, deduplication, and historical tracking patterns.

Master Thesis: Real-Time CNC Monitoring (YOLOv8 Vision System)

- Built a real-time pipeline for data acquisition, preprocessing, and evaluation on an industrial CNC setup.
- Sharpened skills in automation and reproducible workflows while handling large volumes of sensor data.

Certifications

- Azure Data Engineering, End-to-End (ADF, Databricks, ADLS Gen2, Lakehouse, ETL/ELT)
- Data & Data Science (365 Data Science)
- NVIDIA, Fundamentals of Deep Learning

Education

M.Sc. Computer Science, Digital Factory 4.0

Eötvös Loránd University (ELTE)

Thesis: Process Monitoring in Case of Image Processing, CNC Machine (YOLOv8 Vision System)

2025

Hungary

B.Sc. Mathematics & Computing

Kasdi Merbah University

2023

Algeria

Languages

English (C2), French (C1), Arabic (Native), Hungarian (A2, improving), German (Basic)